Trajectory generation for an on-orbit robot manipulator

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Abstract

The problem of trajectory generation for on-orbit robot manipulator like-spacecraft is safety-critical for the success of missions to capture any prescribed target and to berth to another space vehicle. The orbital robot must fulfill the requirements to reach a position with zero relative velocity in close proximity of the target from where the manipulator can grasp a target/spacecraft. The position at zero relative velocity must be inside the robot manipulator workspace. The on-board SW must generate a trajectory envelop and command the manipulator arm (s) so as to accomplish the target grasping without collision. The main difference of such on-board control action with respect to ground manipulators is that in orbit the robot floats and is exited by any control command to move the arm(s) toward a point on the target. When a torque/force is commanded we have reactions on the robot that excites two motion modes on the system. The reaction torque that causes attitude motion (rotational motion) and a reaction force that cause a linear back motion of system instantaneous center of mass keep the orbiting CM in the same position. In this paper we call this configuration as the result of the synchronization of the instantaneous CM with the orbiting CM. A problem arises in such configuration. When the system moves back, it brings together the end effectors. If the trajectory generation does not compensate for this back motion, the robot manipulator fails to reach its target. This paper deals with the problem of trajectory generation which compensate for the robotic system back motion so that the robot arm reaches its target. We do not approach the attitude motion caused by the reaction torque arisen from the joints commanded torque. The reaction torque, if not subject to control actions, causes angular misalignments of the robot manipulator like-spacecraft with respect to the target. Of course this also affects the trajectory generation. In this article we assume that the attitude of both the manipulator like-spacecraft and the target is synchronized. For the case of the linear motion we develop the mathematical model taking into account the instantaneous CM motion to evaluate the amount the robotic system moves back do keep the orbital CM in the same orbital position while the robot arms moves. Such motion is incorporated in the on-board SW trajectory generation module and then the path is tested computationally for precise grasping operations using optimized trajectories.

Introduction

In the scenario of the new paradigm of in-space assembly (ISA), space robot manipulators type-spacecraft are expected to implement activities of RVD/B, target captures and transportation from a rendezvous orbit to another on-orbit construction aiming assembly operation. This paradigm involves fabrication in orbit of modules or components of space vehicles such as space telescopes, space stations, crew modules, etc, and their accessibility in a prescribed rendezvous orbit. The transporter will be robot manipulators-like spacecraft. Such robotic systems shall be as much as possible autonomous and intelligent with secure decision taking capability to prevent collisions, target missing, safe operations requirements related to

maintenance and other on-orbit servicing. The scenario described above shall be implemented in next decades in the context of the New Space Age, with the NASA Artemis program and the Cislunar Gate, a small space station integrated to a series of activities including the return of the man to the Moon for exploration and colonization and later to build a lunar base from where the man shall flight to Mars. The autonomous robot technologies for on-orbit transportation and assembly and for the support in the lunar surface to the astronauts activities will play a critical role for the success of the Moon and Mars explorations. In spite of the scenario resemblance to a science fiction plan, actions for ISA, the Cislunar Gate assembling, and the man return to the Moon are on the way. The plan count on cooperative participation of the main space agencies in the world under NASA coordination and a strong private initiative aiming the Moon exploration. The scenario of the New Space Age will be of cooperation, on-orbit fabrication and vehicle on-orbit assembly aiming cost reduction. The main challenge is to overcome the restrictions imposed by process of building, testing and validating spacecraft/probes on Earth for later launching to space. In the context of the new space age the robot manipulators have being improved to be more and more autonomous with respect teleoperator. To be autonomous mean to have the capability to execute automatically programmed sequence of prescribed tasks. The more complicate the tasks are the more human participation is required to command the robot operations. Dubious interpretations of the tasks may lead the robot to fail. The solution for such problem is artificial intelligence (AI) tool such as computational vision, deep learning our machine learning, for example. The nowadays cutting edge research and robotic application technologies are hardware incorporating AI tools and vision sensors aiming target detail detection and tracking. This will be critical in planetary exploration because in planetary distances the teleoperation of robots is not efficient due to the long time delay of the telecommunication. The main risks involved in manipulators operations in orbit are collisions with target parts or objects surrounding the target, missing target connection points, and failure to grasp objects in space. There are many challenges to be overcome before the new paradigms for space exploration become a common practice in space activities. The research effort is concentrated in the developing high autonomous and intelligent robots. Under the AI umbrella is the computer vision, machine learning, deep learning and on-board software and hardware. Some critical-safety features are intelligence to prevent collisions, missing of targets connection points and assembly ports, interactions between man and machines, EVA operations conducted by autonomous robots, safe communications between robot systems, on-board generations of trajectories for robot manipulator like-spacecraft in microgravity environment. This paper deals with manipulator trajectory generation for save capture of targets connection in orbit. The rendezvous autonomous when Earth control centers support cannot act in real time to interfere in the operations. Close to the Earth the problem has been solved with ground station and astronauts support. However, in planetary missions such as Mars the time delay is to is about 15 minutes and command from Earth is not efficient due to the time delay. NASA is trying to solve the problem by using on-board preprogrammed commands in orbiters. But effective solutions trend point toward autonomous cooperative robot systems for planetary operations.

The center of mass displacement of the robot manipulator during the capture phase may affect the success of the target's grasping because the redistribution of the manipulator links' masses during the robot operation causes a back motion [1] of the whole spacecraft-like manipulator unless a reaction null control is applied. In general the reaction null control counterparts the actions of the joints control system commanded torque to execute the target grasping operation. This control approach may not be effective enough because in such case the center of forces does not coincide with the system center of mass. This fact may cause the robot to oscillate about the center of forces, impacting the attitude motion. The reaction null technique would have to take into account the attitude stabilization in the

transient phase when the manipulator links are moving toward the target. The approach used in this paper is to tolerate the natural rearrangement of the back motion of the spacecraft-like manipulator, accommodating the instantaneous center of mass to the orbiting CM, that is, synchronizing the instantaneous CM with the orbital motion. In this case, however, the amount of back displacement must be accounted for in the trajectory planning to avoiding the capture point missing by the end effectors.

Consider the following physical model of a manipulator like-spacecraft consisting of an orbital mobile base and three revolute joints as shown in figure 1, where

θ_i , $i = 1,2$ are the joint angles;

$\dot{\theta}_i$, $i = 1,2$ are the joint angular rates ;

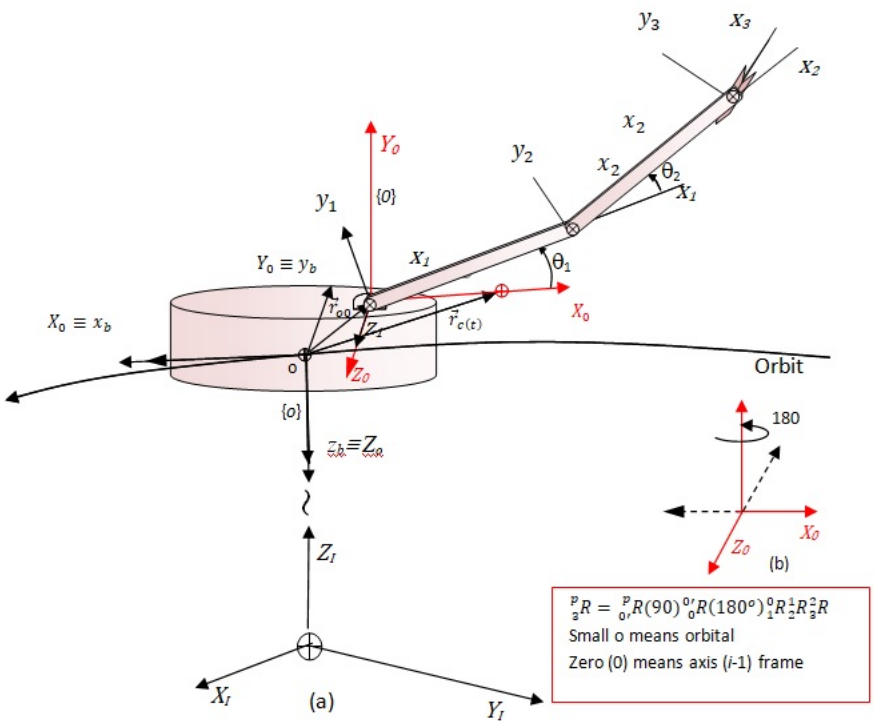

Figure 1 Space robot manipulator like-spacecraft system

$X_oY_oZ_o$ is the orbital frame, RPY (Roll, Pitch, Yaw), also called LVLH (Local Vertical, Local Horizontal) [2,3];

$X_0Y_0Z_0$ is the frame for $\{i-1\}$ for $i = 1$. It is fixed with respect to the platform and moves in attitude with the manipulator like-spacecraft

$x_py_pz_p$ is the body fixed frame .The equations of motion are written in this system.

$X_IY_IZ_I$ is the Earth-Centered-Inertial reference frame [3];

CM is the system center of mass and it appears in two different positions. One when the robot arm is folded on the platform and is out of service. The second, refers to the CM(t) , when the manipulator is on and moving toward the target connection point. The position vectors for both CM positions are given by $\vec{r}_{o0}$ and $\vec{r}_c(t)$. Note that when the manipulator is on and moving toward the target connection point it occurs a system mass reconfiguration changing

the position of the CM. The displacement changes with respect to time during the transient of the target capture.
Frame $x_b y_b z_b$ is assumed to be aligned with the frame $x_o y_o z_o$ when the manipulator links start moving. In this configuration of the orbital and body frame alignments the attitude is zero.
In figure 1, the origin of $\{0\}$ relative to {b} is defined by the point $\left(\frac{d}{4},\ 0, -\frac{h}{2}\right)$, where d and h are the diameter and the height of the manipulator base.
By using the definition of the system center of mass we find the components of the position vector of the CM as:

$$x_c = -\frac{d}{4} + \frac{\ell}{6}[5c\theta_1 + 3c(\theta_1 + \theta_2) + c(\theta_1 + \theta_2)] \quad (1)$$

$$z_c = -\frac{h}{2} + \frac{\ell}{6}[5s\theta_1 + 3s(\theta_1 + \theta_2) + s(\theta_1 + \theta_2)]$$

$$y_c = 0$$

where the link masses are assumed to be equal, so that

$$m = \sum_{i=1}^{3} m_i = 3m$$

$$M = M_b + m$$

ℓ is the length of each link

c and *s* means cosine e sine respectively

$\theta_i = \theta_i(t)$, the joint angles

t is the time.

The ideia is to implement the computer simulation of the mathematical model derived for the equations of dynamics, considering the orbital and attitude motions. The orbital motion is represented for the Hill's equation while the attitude motion involves the rotational motion of the manipulator as a spacecraft and the joint motion in the transient phase when the manipulator moves to the target connection point. The results of the numerical integration shall provide as outputs the joint angles during the referred transient phase. This outputs feeds the equation of the CM so that during the transient phase the distance the center of moved can be computed and used to compensate the trajectory generation so as the manipulator arm can berth with the target by connecting successfully.

In figure 1, the vector position from the origin of B frame to the origin of frame 1 (link 1) defines the point coordinates which is assumed to be at the point $\left(\frac{d}{4},\ 0, \frac{h}{2}\right)$, where d and h are the diameter and the height of the manipulator base.

By using the definition of the system center of mass we find the components of the position vector of the CM as:

In general, books approaches ground robot manipulators with a fixed base, as in Craig's [4], for example. For orbital applications the robot base cannot be considered as fixed. Rather the robot base floats in the microgravity environment and experiences three types of motions, rotation about its center of mass (attitude motion), revolution of its center of mass about the Earth (orbital motion), and a linear motion of its center of mass in the contrary direction of the end effectors motion during the transient phase of target capture. These motions apply for the complete robot manipulator like spacecraft. As a spacecraft the attitude motion carry out also the arm's links be them out or in operation. When in operation mode the attitude motion must be synchronized to the target attitude that is, the relative attitude must be zero and stable to avoid the manipulator missing the target contact point. This is not a trivial task in practice because sensor technology must be used to detect e track the target contact point and control actuators must be selected to control the relative motion to avoid failure of manipulator to accomplish the connection with the target connection point. inform the on-board control system so that the A solution for the complex dynamics of a space manipulator in the microgravity environment is the use of active control with appropriate actuators and sensors to stabilize the robot attitude. Up to date there is not a safe solution without the use of telecontrol or telerobotics. In this way we take the advantage of the small displacement of the center of mass motion during the transient for the capture of the target and include it in the trajectory generation. In this way we avoid the target point missing in the action of capture.

By following the D-H convention the parameters associate to the manipulator shown in figure 1 are shown in table 1 as

D-H Parameter				
i	α_{i-1}	a_{i-1}	d_{i-1}	θ_i
1	*0*	*0*	*0*	θ_1
2	0	l_1	*0*	θ_2

Table 1 D-H parameters for the manipulator shown in figure 1

The homogenous transform to frame {0} from frame {3} is given by

$$ {}^{0}_{1}T\,{}^{1}_{2}T\,{}^{2}_{3}T = \begin{bmatrix} c\theta_1 & -s\theta_1 & 0 & 0 \\ s\theta_1 & c\theta_1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} c\theta_2 & -s\theta_2 & 0 & \ell_1 \\ s\theta_2 & c\theta_2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{2} $$

The rotation matrix from {0} to the platform frame {p} is

$$ {}^{x_b y_b z_b}_{x_0' y_0' z_0'}\begin{bmatrix} 1 & 0 & 0 \\ 0 & c90 & -s90 \\ 0 & s90 & c90 \end{bmatrix} {}^{x_0' y_0' z_0'}_{x_0 y_0 z_0}\begin{bmatrix} c(180) & 0 & sin180 \\ 0 & 1 & 0 \\ -sin180 & 0 & c(180 \end{bmatrix} = {}^{x_p y_p z_p}_{x_0 y_0 z_0}\begin{bmatrix} -1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} $$

$$ {}^{b}_{0}\boldsymbol{R} = \left[\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix}\begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix} = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}\right] $$

So the transform to {0} from (3} is

$$ {}^{0}_{2}T = \begin{bmatrix} cos(\sum_1^2 \theta_i) & -sin(\sum_1^2 \theta_i) & 0 \\ sin(\sum_1^2 \theta_i) & cos(\sum_1^2 \theta_i) & 0 \\ 0 & 0 & 1 \end{bmatrix} \tag{3} $$

and from {3} to {p} is

$$ {}^{p}_{2}T = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} cos(\sum_1^2 \theta_i) & -sin(\sum_1^2 \theta_i) & 0 \\ sin(\sum_1^2 \theta_i) & cos(\sum_1^2 \theta_i) & 0 \\ 0 & 0 & 1 \end{bmatrix} \tag{4} $$

Note that {p} is fixed on in the platform of the manipulator like-spacecraft that is free to move under attitude perturbations in the micro gravity environment, unless there is an attitude control system is design stabilize it. In this article we consider two attitude control systems so as the platform may be controlled and stabilized in attitude.

By following the Craig's approach based on the D-H notation the kinematics equation can be written as

$$ {}^{1}\Omega_1 = \begin{bmatrix} 0 \\ 0 \\ \dot{\theta}_1 \end{bmatrix}; \; {}^{2}\Omega_2 = \begin{bmatrix} 0 \\ 0 \\ \dot{\theta}_1 + \dot{\theta}_2 \end{bmatrix}; \tag{5} $$

$$ {}^{1}\upsilon_1 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \tag{6} $$

$$ {}^{2}\upsilon_2 = \begin{bmatrix} c\theta_2 & s\theta_2 & 0 \\ -s\theta_2 & c\theta_2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ \ell_1\dot{\theta}_1 \\ 0 \end{bmatrix} = \begin{bmatrix} \ell_1 s\theta_2\dot{\theta}_1 \\ \ell_1 c\theta_2\dot{\theta}_1 \\ 0 \end{bmatrix} \tag{7} $$

$$ {}^{3}\upsilon_3 = \begin{bmatrix} \ell_1 s\theta_2\dot{\theta}_1 \\ \ell_1 c\theta_2\dot{\theta}_1 + \ell_2(\dot{\theta}_1 + \dot{\theta}_2) \\ 0 \end{bmatrix} \tag{8} $$

In this work the {p} is used to write the equation of motion. Then it is necessary to find the transform frm {3} to {p}.

$$ {}^{p}\upsilon_3 = {}^{p}_{3}T = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} cos(\theta_1 + \theta_2) & -sin(\theta_1 + \theta_2) & 0 \\ sin(\theta_1 + \theta_2) & cos(\theta_1 + \theta_2) & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \ell_1 s\theta_2\dot{\theta}_1 \\ \ell_1 c\theta_2\dot{\theta}_1 + \ell_2(\dot{\theta}_1 + \dot{\theta}_2) \\ 0 \end{bmatrix} \tag{9} $$

$$ {}^{p}\upsilon_3 = \begin{bmatrix} -\cos(\theta_1 + \theta_1) & sin(\theta_1 + \theta_2) & 0 \\ 0 & 0 & 1 \\ sin(\theta_1 + \theta_2) & cos(\theta_1 + \theta_2) & 0 \end{bmatrix} \begin{Bmatrix} \dot{\theta}_1 \\ \dot{\theta}_2 \\ 0 \end{Bmatrix} = $$

$$ \begin{bmatrix} -1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \begin{Bmatrix} -\cos(\theta_1 + \theta_1)\dot{\theta}_1 + sin(\theta_1 + \theta_2)\dot{\theta}_2 \\ 0 \\ sin(\theta_1 + \theta_2)\dot{\theta}_1 + cos(\theta_1 + \theta_2)\dot{\theta}_2 \end{Bmatrix} = \begin{Bmatrix} \cos(\theta_1 + \theta_1)\dot{\theta}_1 - sin(\theta_1 + \theta_2)\dot{\theta}_2 \\ sin(\theta_1 + \theta_2)\dot{\theta}_1 + cos(\theta_1 + \theta_2)\dot{\theta}_2 \\ 0 \end{Bmatrix} \tag{10} $$

This transformation leads the frame to frame $\{0\}$ from $\{o\}$ that from the convention os this paper approach for relationshops of frames, is $\{0\} \leftarrow \{o\}$

We must check the Jacobian for singularity. This can be done by computing the determinant of the $\mathcal{J}$.

In a similar manner we can treat the angular velocities Ω wich are in fact given by the time derivatives of the θ_i.

Equations of Dynamics

The dynamics equations for the manipulator like-spacecraft are sometimes called as the Euler modified equations. In spite of the platform and links be considered rigid, the system is not rigid because robot manipulator links are movable with respect to the each other and also with respect of the rigid platform. Rather, the system includes internal degrees-of-freedom given by the joint angles. The resulting Euler's equation involves coupling between the system rigid body motion and the angle and rates of the manipulator joints. By using Lagrange's energy formulation to derive the Euler modified equations seems appropriate to obtain the modified Euler's equations. To approach the formulation by Lagrange's formulation the energy equations for complete system shall be derived and then the Lagrange's formula for quasi coordinates is used in conjunction with the Lagrange's formula for generalized coordinates. Of course we can write the vector $\{\omega\}$ as a function of the attitude angles and then use only the Lagrange's formula for generalized coordinates. However by using both formulations for quasi-coordinates and for generalized coordinates the structure of the Euler's equations are preserved.

The kinetics energy for the system is

$$T = \sum_{i=1}^{3} T_i + T_p \tag{11}$$

Where i refers to the arm links and p to the base (platform) of the manipulator like-spacecraft. This approach for the derivation of the equation of motion is presented in [5]. In addition to the equations of the dynamics it is also included in the model the system variable center of mass (CM), denoted here as $r_{c.}(\theta_i(t))$. This approach is consistent with the reconfiguration of link's masses m_i with respect to the platform, when the manipulator is on operation mode. Rather the mass reconfiguration and consequent displacement of the system CM have impact on the dynamics of the manipulator like-spacecraft. The assumption of a time varying position of the system center of mass is valid only in the transient phase when the robot arm moves toward the target contact point for berthing operation. Once the contact is effective a new dynamics system forms with links locked. In case of space transportation of the target by the manipulator the arms can be commanded to move the target to somewhere. This phase of the operation is out of the scope of this work and will be a subject of another research paper. Details of the derivation are shown for the instantaneous manipulator center of mass and for the trajectory generation through polynomials. but some algebraic manipulation is presented for the typical linear transformation involving robot manipulators supported by a non fixed platforms operating in microgravity environment. The explicit equations of kinetics energies written with respect to the platform frame, considering the manipulator in operational mode are

$$T_p = \frac{1}{2}\{\omega\}^T [I_p]\{\omega\} \tag{12}$$

$$T_i = \frac{1}{2} m_i {v_{ci}}^2 + \frac{1}{2}\{\omega\}^T {}^p_i R^T [I_i] {}^0_i R\{\omega\} \tag{13}$$

$$T_1 = \frac{1}{2}\{\omega + \dot{\Theta}_1\}^T {}^p_0R^T {}^0_1R^T [I_1] {}^p_0R^T {}^0_1R^T \{\omega + \dot{\Theta}_1\} + \frac{1}{2} m_{c1} \ell_{c1}^2 \dot{\theta}_1^2 \tag{14}$$

$$T_2 = \frac{1}{2}\{\omega + \dot{\Theta}_2\}^T \left({}^p_0R^T {}^0_1R^T {}^1_2R\right)^T [I_2] {}^p_0R {}^0_1R {}^1_2R \{\omega + \dot{\Theta}_2\} + \frac{1}{2} m_2 \ell_{c2}^2 \dot{\theta}_2^2 \tag{15}$$

Where $\dot{\Theta}_1 = \begin{Bmatrix} 0 \\ 0 \\ \dot{\theta}_1 \end{Bmatrix}$; $\dot{\Theta}_2 = \begin{Bmatrix} 0 \\ 0 \\ \dot{\theta}_1 + \dot{\theta}_2 \end{Bmatrix}$; ; $\{\omega\} = \begin{Bmatrix} \omega_{x_p} \\ \omega_{y_p} \\ \omega_{z_p} \end{Bmatrix}$

$$\ell_{c1}^2 = \left(\frac{\ell_1}{2}\right)^2 ; \ \ell_{c2}^2 = \ell_1^2 + \left(\frac{\ell_2}{2}\right)^2 + \ell_1 \ell_2 \mathrm{c}\, \theta_2;$$

$$\ell_{c3}^2 = \ell_1^2 + \ell_2^2 + \ell_3^2 + 2\ell_2\ell_3 cos\theta_3 + 2\ell_1(\ell_2^2 + \ell_3^2 + 2\ell_2\ell_3 cos\theta_3) cos\theta_2$$

Assuming the principal axes of the platform and of each link the inertia matrices can be written as

$$[I_p] = \begin{bmatrix} I_{xp} & 0 & o \\ 0 & I_{yp} & 0 \\ 0 & 0 & I_{zp} \end{bmatrix}; \quad {}^{c_1}[I_1] = \begin{bmatrix} I_{x1} & 0 & o \\ 0 & I_{y1} & 0 \\ 0 & 0 & I_{z1} \end{bmatrix}; \quad {}^{c_2}[I_2] = \begin{bmatrix} I_{x2} & 0 & o \\ 0 & I_{y2} & 0 \\ 0 & 0 & I_{z2} \end{bmatrix}; \tag{16}$$

Where I_i is inertia matrix for the link i with respect to its center of mass position.

No potential energy is considered but only the rotational motion for the system so that the use of Lagrangian formulation for quasi-coordinates and the Lagrangian formulation [6] for generalized coordinates is appropriate for the derivation of the equations of motion. The Langrage function is explicitly function of the quasi-coordinates and the generalized coordinates associate to the robot arm degrees-of-freedom. In a compact form:

$$L = T = \frac{1}{2}\{\omega\}^T [I_p]\{\omega\} + \sum_{i=1}^{3} T_i \tag{17}$$

$$T_i = \frac{1}{2} m_i {v_{ci}}^2 + \frac{1}{2}\{\omega\}^{T\, J_i}[I_i]\{\omega\} \tag{18}$$

$$T_1 = \frac{1}{2} I_{x1}\omega_x^2 + I_y\omega_y^2 + I_z(\omega_z + \dot{\theta}_1)^2 \tag{19}$$

$$T_2 = \frac{1}{2} I_{x_2}\omega_x^2 + I_y\omega_{y_2}^2 + I_{z_2}(\omega_z + \dot{\theta}_1 + \dot{\theta}_2)^2 \tag{20}$$

$$T_{T=} \frac{1}{2}\{\omega\}^T [I_p]\{\omega\} + \frac{1}{2} I_{x1}\omega_x^2 + I_y\omega_y^2 + I_z(\omega_z + \dot{\theta}_1)^2 + +I_{z_3}(\omega_z + \dot{\theta}_1 + \dot{\theta}_2 + \dot{\theta}_2)^2 \tag{21}$$

The equation are written with respect to the platform frame. As the inertia matrices are computed in separate body frames, that is, the platform and links, rotation matrices and the parallel axis theorem is applied to represent all the inertia matrices in the platform system. The formula to derive the Euler modified equation and equation of dynamics for the manipulator joint motion is, respectively

$$\frac{d}{dt}\left\{\frac{\partial T}{\partial \omega}\right\} + [\widetilde{\omega}]\left\{\frac{\partial T}{\partial \omega}\right\} = \{N_g + N_m\} + \{N_c\} \tag{22}$$

$$\frac{d}{dt}\frac{\partial T}{\partial \dot{\theta}_i} + \left\{\frac{\partial T}{\partial \theta_i}\right\} = N_{\Theta_i} \tag{23}$$

Where

T is total kinetic energy, $\{Ng + Nm\}$ is the perturbation due to the gravity-gradient and the residual torque related to the residual magnetic field inside the manipulator like-spacecraft. $\{Nc\}$ is the control torque by using reaction wheels, $[\widetilde{\omega}]$ is a skew symmetric matrix in the components of the vector $\{\omega\}$,and N_{Θ_i} torques commanded to the manipulator joints, θ_i.

$$\left\{\frac{\partial T}{\partial \omega}\right\} = \begin{bmatrix} J_{11} & -J_{12} & 0 \\ -J12 & J_{22} & 0 \\ 0 & 0 & J_{33} \end{bmatrix} \begin{Bmatrix} \omega_x \\ \omega_y \\ \omega_z \end{Bmatrix} + \begin{Bmatrix} 0 \\ 0 \\ I_{z1}\dot{\theta}_1 + I_{z2}(\dot{\theta}_1 + \dot{\theta}_2) \end{Bmatrix} \tag{24}$$

where

$J_{11} = I_{xp} + I_{x1}\cos(\theta_1) + I_{x2}\cos(\theta_1 + \theta_1)$

$J_{12} = (m_1\ell_{c1}^2 + I_{y1})sin\theta_1 + (m_2\ell_{c2}^2 + I_{y2})sin(\theta_1 + \theta_1)$

$J_{21} = J_{12}$

$J_{22} = I_p + (m_1\ell_{c1}^2 + I_{y1})cos\theta_1 + (m_2\ell_{c2}^2 + I_{y2})cos(\theta_1 + \theta_1)$

$J_{33} = I_{zp} + I_{z1} + I_{z2} + m_1\ell_{c1}^2 + m_2\ell_{c2}^2$

$$\frac{d}{dt}\left\{\frac{\partial T}{\partial \omega}\right\} + \left\{\frac{\partial T}{\partial \omega}\right\} = \begin{bmatrix} J_{11} & -J_{12} & 0 \\ -J12 & J_{22} & 0 \\ 0 & 0 & J_{33} \end{bmatrix} \begin{Bmatrix} \dot{\omega}_x \\ \dot{\omega}_y \\ \dot{\omega}_z \end{Bmatrix} + \begin{bmatrix} 0 & -\omega_z & \omega_y \\ \omega_z & 0 & -\omega_x \\ -\omega_y & \omega_x & 0 \end{bmatrix} \begin{bmatrix} J_{11} & -J_{12} & 0 \\ -J12 & J_{22} & 0 \\ 0 & 0 & J_{33} \end{bmatrix} \begin{Bmatrix} \omega_x \\ \omega_y \\ \omega_z \end{Bmatrix} \tag{25}$$

$$+ \begin{Bmatrix} 0 \\ 0 \\ I_{z1}\ddot{\theta}_1 + I_{z2}(\ddot{\theta}_1 + \ddot{\theta}_2) \end{Bmatrix} = \left\{\{N_g + N_m\} + \{N_c\}\right\}$$

The equations for the generalized coordinates associate to the manipulator degrees-of-freedom are

$$\left(\frac{\partial T}{\partial \dot{\theta}}\right) = \left(m_1(\ell_1^2 + \ell_{c1}^2) + m_2(\ell_{c1}^2 + 2\ell_{c1}\ell_{c2}cos\theta_2 + 2\ell_{c2}^2 + I_{z1} + I_{z2})\right)\dot{\theta}_1 + \left(I_{z2} + m_2(2\ell_{c2}^2 + \ell_{c1}\ell_{c2}cos\theta_2)\right)\dot{\theta}_2 + \left(I_{z1} + I_{z2} + 2(m_1\ell_{c1}^2 + m_2\ell_{c2}^2)\right)\omega_z$$

$c_{t1} = \left(m_1(\ell_1^2 + \ell_{c1}^2) + m_2(\ell_{c1}^2 + 2\ell_{c1}\ell_{c2}cos\theta_2 + 2\ell_{c2} + I_{z1} + I_{z2})\right)$

$c_{t2} = \left(I_{z2} + m_2(2\ell_{c2}^2 + \ell_{c1}\ell_{c2}cos\theta_2)\right)$

$c_z = \left(I_{z1} + I_{z2} + 2(m_1\ell_{c1}^2 + m_2\ell_{c2}^2)\right)$

$$\frac{\partial T}{\partial \dot{\theta}} = c_{t1}\dot{\theta}_1 + c_{t2}\dot{\theta}_2 + c_z\omega_z$$

$$\frac{d}{dt}\left(\frac{\partial T}{\partial \dot{\theta}_1}\right) = c_{t1}\ddot{\theta}_1 + c_{t2}\ddot{\theta}_2 + c_z\dot{\omega}_z - 2\ell_{c1}\ell_{c2}sin\theta_2(\dot{\theta}_1\dot{\theta}_2) - \ell_{c1}\ell_{c2}sin\theta_2(\dot{\theta}_2^2)$$

$$\frac{\partial T}{\partial \theta} = \frac{1}{2}\left((I_{x1}-I_{y1})\omega_y^2 + (I_{y1}-I_{x1})\omega_x^2\right)sin(2\theta_1) + \frac{1}{2}\left((I_{x2}-I_{y2})\omega_y^2 + (I_{y2}-I_{x2})\omega_x^2\right)sin(2\theta_1+2\theta_2)$$
$$+\left((I_{y1}-I_{x1})cos(2\theta_1) + (I_{y2}-I_{x2})cos(2\theta_1+2\theta_2)\right)\omega_x\omega_y$$

$$\frac{d}{dt}\left(\frac{\partial T}{\partial \dot{\theta}_1}\right) - \frac{\partial T}{\partial \theta_1} = c_{t1}\ddot{\theta}_1 + c_{t2}\ddot{\theta}_2 + c_z\dot{\omega}_z - 2\ell_{c1}\ell_{c2}sin\theta_2(\dot{\theta}_1\dot{\theta}_2) - \ell_{c1}\ell_{c2}sin\theta_2(\dot{\theta}_2^2)$$
$$-\frac{1}{2}\left((I_{x1}-I_{y1})\omega_y^2 + (I_{y1}-I_{x1})\omega_x^2\right)sin(2\theta_1)$$
$$+\frac{1}{2}\left((I_{x2}-I_{y2})\omega_y^2 + (I_{y2}-I_{x2})\omega_x^2\right)sin(2\theta_1+2\theta_2)$$
$$+\left((I_{y1}-I_{x1})cos(2\theta_1) + (I_{y2}-I_{x2})cos(2\theta_1+2\theta_2)\right)\omega_x\omega_y = N_{\Theta_i} \tag{26}$$

$$\frac{\partial T}{\partial \dot{\theta}_2} = (I_{z2} + 2m_2\ell_{c2}^2)\dot{\theta}_2 + (I_{z2} + m_2\ell_{c1}cos(\theta_2) + 2m_2\ell_{c2}^2)\dot{\theta}_1 + (I_{z2} + m_2\ell_{c2}^2)\omega_z$$

$$\frac{d}{dt}\left(\frac{\partial T}{\partial \dot{\theta}_2}\right) = (I_{z2} + 2m_2\ell_{c2}^2)\ddot{\theta}_2 + (I_{z2} + m_2\ell_{c1}cos(\theta_2) + 2m_2\ell_{c2}^2)\ddot{\theta}_1 + (I_{z2} + m_2\ell_{c2}^2)\dot{\omega}_z$$
$$- m_2\ell_{c1}sin(\theta_2)\dot{\theta}_1\dot{\theta}_2$$

$$\frac{\partial T}{\partial \theta_2} = \frac{1}{2}(I_{y2}-I_{x2})\left((\omega_x^2-\omega_y^2)sin(2\theta_1+2\theta_2) + 2\omega_x\omega_y cos(2\theta_1+2\theta_2)\right)$$

$$\frac{d}{dt}\left(\frac{\partial T}{\partial \dot{\theta}_2}\right) - \frac{\partial T}{\partial \theta_2} = (I_{z2} + 2m_2\ell_{c2}^2)\ddot{\theta}_2 + (I_{z2} + m_2\ell_{c1}cos(\theta_2) + 2m_2\ell_{c2}^2)\ddot{\theta}_1 + (I_{z2} + m_2\ell_{c2}^2)\dot{\omega}_z$$
$$- m_2\ell_{c1}sin(\theta_2)\dot{\theta}_1\dot{\theta}_2$$
$$-\frac{1}{2}(I_{y2}-I_{x2})\left((\omega_x^2-\omega_y^2)sin(2\theta_1+2\theta_2) + 2\omega_x\omega_y cos(2\theta_1+2\theta_2)\right) \tag{27}$$

These are the sets of the equations of motion.

The kinematics differential equations are written in terms of quaternion to avoid the typical singularities associated to the Euler angles formulation. However the quaternion formulation is used in conjunction with the Euler angles. The quaternion are used for the sake of computer simulations and then the a transformation for quaternion to Euler angles is obtained algebraically to plot the graphics and to show up more clearly the physical meanings of the results. The quaternion formulation for the attitude motion dynamics analysis is not shown in details here but the following reference [7] are provided for those that are not familiar with the formulation. The Transformation to the platform frame to the orbital frame in terms of quaternion is

$$\begin{bmatrix} 1-2(q_2^2+q_3^2) & 2(q_1q_2+q_3q_4) & 2(q_1q_3-q_2q_4) \\ 2(q_1q_2-q_3q_4) & 1-2(q_1^2+q_3^2) & 2(q_2q_3+q_1q_4) \\ 2(q_1q_3+q_2q_4) & 2(q_2q_3-q_1q_4) & 1-2(q_1^2+q_2^2) \end{bmatrix} \tag{28}$$

Now consider the transformation obtained by the LVLH frame (sequence of rotations 3-2-1 or as Bong We notation, $C_X(\varphi) \longleftarrow C_Y(\theta) \longleftarrow C_Z(\psi)$

$$\begin{bmatrix} c\theta c\varphi & c\theta s\varphi & -s\theta \\ s\psi s\theta c\varphi - c\psi s\varphi & s\psi s\theta s\varphi + c\psi c\phi & s\psi c\theta \\ c\psi s\theta c\varphi + s\psi s\varphi & c\psi s\theta s\phi - s\psi c\phi & c\psi c\theta \end{bmatrix} \tag{29}$$

As both transformations must be equal, the problem of finding the angles in terms of quaternion can be solved by comparing the elements to elements of both matrices. The transformation as a function of the attitude angles is obtained by the transformation to platform frame from the LVLH frame. This yields

$$\psi = tg^{-1}\left(\frac{2(q_1q_2 + q_3q_4)}{1 - 2(q_1^2 + q_2^2)}\right)$$

$$\theta = -sin^{-1}\big(2(q_1q_3 - q_2q_4)\big) \tag{30}$$

$$\varphi = tg^{-1}\left(\frac{2(q_2q_3 + q_1q_4)}{1 - 2(q_2^2 + q_3^2)}\right)$$

The quaternion kinematics differential equations can be written as

$$\begin{Bmatrix} \dot{q}_1 \\ \dot{q}_2 \\ \dot{q}_3 \\ \acute{q}_4 \end{Bmatrix} = \begin{bmatrix} 0 & \omega_z & -\omega_y + n & \omega_x \\ -\omega_z & 0 & \omega_x & \omega_y + n \\ -\omega_y + n & -\omega_x & 0 & \omega_z \\ -\omega_x & -\omega_2 - n & -\omega_z & 0 \end{bmatrix} \begin{Bmatrix} q_1 \\ q_2 \\ q_3 \\ q_4 \end{Bmatrix} \tag{31}$$

Where n is the orbital angular rate about Earth for orbits approximately circular as it is assumed in this work. The orbital angular rate n is given by

$$n = \sqrt{\frac{G(M+m)}{a^3}}. \tag{32}$$

For satellites orbiting the Earth the small mass m compared to Earth mass M can be neglected.

In the Euler angles this kinematics differential can be written as

In the Euler Angles, the kinematics differential equations can be written in matrix format as

$$\begin{Bmatrix} \omega_X \\ \omega_Y \\ \omega_Z \end{Bmatrix} = \begin{bmatrix} 1 & 0 & -S\theta \\ 0 & c\varphi & s\varphi c\theta \\ 0 & -s\varphi & c\varphi c\theta \end{bmatrix} \begin{Bmatrix} \dot{\varphi} \\ \dot{\theta} \\ \dot{\psi} \end{Bmatrix} - n \begin{Bmatrix} c\theta s\psi \\ s\varphi s\theta s\psi + c\varphi c\psi \\ c\varphi s\theta s\psi - s\varphi c\psi \end{Bmatrix} \tag{33}$$

The kinematics differential equations can be written as

$$\begin{Bmatrix} \dot{\varphi} \\ \dot{\theta} \\ \dot{\psi} \end{Bmatrix} = \frac{1}{c\theta} \begin{bmatrix} c\theta & s\varphi s\theta & c\varphi s\theta \\ 0 & c\varphi c\theta & -s\varphi c\theta \\ 0 & s\varphi & c\varphi \end{bmatrix} \begin{Bmatrix} \omega_X \\ \omega_Y \\ \omega_Z \end{Bmatrix} + \frac{n}{c\theta} \begin{Bmatrix} s\psi \\ c\theta c\psi \\ s\theta s\psi \end{Bmatrix} \tag{34}$$

This equation is for a spacecraft orbiting the Earth with the rate n. Note that singularities will occur in $c\theta$ approximate to zero. The quaternion formulation overcome this problem. The kinematics differential equations as presented here if for the LVLH frame and sequence of rotations 3-2-1 or $C_X(\varphi) \longleftarrow C_Y(\theta) \longleftarrow C_Z(\psi)$

The Control Design

The LQR technique can be state as.

Minimize a cost function given

$$J=\int_{0}^{\propto} \{x\}^T [Q]\{x\}+\{u\}^T [R]\{u\} dt \qquad 35$$

Subject to

$$\{\dot{x}\}=[A]\{x\}+[B]\{u\} \qquad 36$$

$$\{u\}=-[k]\{x\}$$

where the gain is given by

$$[k]=R^{-1}B^T P$$

$\{\dot{x}\} = [A]\{x\} + B\{u\}$ is the dynamics written in 2n equation in the state vector form

{*u*} is the state feedback control law that result of the quadratic performance index optimization subjected to the state equations. [A] and [B] are the plant and input matrices, respectively. [Q] is a positive semi definite state weighting matrix, and [R] is the control weighting matrix that has to be positive definite. Q is 2 n×2 n and R is a p×p matrix. This technique writing the equations of dynamics in the state vector format also requires the solution of the Riccati algebraic to obtain [*k*]:

$$[A]^T [P]+[P][A]+[Q]-[P][B][R]^{-1}[B]^T [P]=0 \qquad 37$$

The PID ControlTtechnique

This approach is selected for the não linear dynamcis analysis [8]. The PID combines an integral action, an on the other hand, gives a controller output that is proportional to the accumulated error, which implies that it is a slow reaction control mode. And finally, the derivative action acts on the rate of change of the control error. The mathematical equation of a PID controller can be written as

$$N_c=k_p\theta_e+k_d\dot{\theta}_e+k_i \int \theta_e dt \qquad 38$$

A simple gain selection strategy based on the Ziegler–Nichols rule has been implemented. The proportional and derivative gains are obtained by taking into account the inertia of the system, the pulsation and the desired control damping. The gains are obtained by

$$k_p = -diag(I)\omega$$

$$k_d = -2diag(I)\omega\zeta$$

$$k_i = \frac{k_p^2}{2k_d}$$

where I, ω, and ζ are the inertia matrix, pulsation and damping,respectively.

Computational Simulations and Results

The MATLAB SW package was used to extensivel to obtain the equation of motion. The SW symbolic tool box is apropriate to implement all the algebraic manipulations to generate the transformation matrices, compute the jacobians and check for singuraties. The SW package is lused to integrates the esquations of motion, implement the control law and gnerate the garphics.

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